

TSPI AI Engine — Project Status & Update

Prepared for the TSPI clinical / domain-expert team · Last updated: 23 Aug 2026

This document summarises where the TSPI AI Engine stands today: what is built and tested, what remains, and — most importantly — **what we now need from your team to keep the project moving**. The engine's core is complete; the remaining timeline depends largely on a few clinical data approvals that only your team can provide.

Update — incorporating your 5 Aug Final Review. Your final ruling on severity is now implemented: **data completeness no longer reduces an axis's severity** (NSS reworked to v0.3 — a severe finding stays severe, with reliability shown separately). **CELA is recorded as approved** (TSPI-NET-0120 → A34 / D10), and the A10 sub-axes and the nine Restoration Steps are locked to your canonical lists. Your review also added a pre-pilot governance layer (see §7) and corrected one assumption: the 25% network match depends on a new **Module-to-Network Claim Registry**, not the axis crosswalk alone.

1. What TSPI is (one paragraph)

TSPI is the registry-driven, deterministic AI "brain" behind MiHealth. Reasoning flows **Clinical Evidence → Evidence Integration → 3 Keys → 9 Restoration Steps → 12 Domains → 39 Biological Axes → 180 Living Networks → Module ranking → Physician approval → Reports**. The **39 Axes are the central diagnostic and module-matching layer**; the 180 Networks *explain and validate* mechanisms — they never replace axis matching. The AI only summarises and translates; it never invents an axis score, network identity, module, research result, or outcome. Frozen ontology: `framework_version = 39-axis-master-260715`.

2. Constitutional rules (permanent, always enforced)

Deterministic engine · official-registry-only resolution · **missing data → NOT_ASSESSED, never 0, never a default of 50** · confidence/completeness affect *reliability, not severity* · Evidence Status ≠ Evidence Grade ≠ Confidence ≠ Completeness · a hypothesis never scores · **safety is a categorical gate, not a number** · red-flag screening runs first · **learning is propose-only** (a clinician always approves) · one immutable analysis record per run · the system fails closed when a registry is missing or unapproved · physician approval before any patient release.

3. Built & tested — engine core ✓ (82 automated tests passing)

Area	State
Full pipeline (normalise → axis-map → severity → module-match → safety → compose → validate)	✓
Evidence model — 4 dimensions + biological uncertainty, explicit NOT_ASSESSED	✓
Red-flag safety screening — deterministic, runs first, critical-value layer	✓
Phenotype engine + named Phenotype Registry (Option A, candidate list)	✓
Network Severity Score v0.3 (completeness never reduces severity) + Module Match v1.0	✓
Legacy → current axis converter (auto-applies safe cases, quarantines the rest)	✓
Learning — propose-only, versioned patient model	✓
Marker Registry — value source vs direction rule, curve types, context markers	✓
Canonical records locked — CELA (TSPI-NET-0120 →A34/D10), A10 sub-axes, 9 Restoration Steps	✓
All refinements from your 25 Jul, 31 Jul and 5 Aug guidance	✓ (see §5)
Doctor's-chat integration (MCP) — connect the AI Brain to an AI chat client	✓ (see §8)
Approve and update a plan — structured, reason-coded physician edits	✓
Access control — authentication, roles (patient / clinic-staff / clinician / reviewer / auditor), audit trail	✓ (see §8)

4. Remaining work — and what it depends on

The work still ahead falls into three groups. **Group 4a we can do on our own. Groups 4b and 4c depend on your team and are on the critical path** — every item there blocks downstream work until it is provided.

4a. What we are building now (no dependency on you)

We continue in parallel while awaiting your approvals:

- **Network layer integration** — loading the 180 networks into the engine and wiring the mechanistic- validation step (today the network score is a safe placeholder marked "not assessed").
- **Report layer** — the Physician, Patient, and Multi-Omics reports, **Thai-first bilingual**, all reading from one immutable analysis record, with a cross-report consistency check and physician e-signature.
- **Minimum-evidence gate, governance controls, and the remaining registry loaders** (lab cut-offs, research claims, restoration-step mapping) — ready to receive your clinical content the moment it arrives.

4b. ▲ What we need from you — URGENT, on the critical path

These are the items that currently **hold up clinically usable output**. Until they are provided, the engine can reason and rank internally, but its recommendations **cannot be released for clinical use**. Each week of delay pushes back the physician pilot and, after it, patient-facing reports.

#	What we need	Why it blocks everything downstream	If delayed
1	Approved Module Registry (Blocker 1) — the module list mapped to current A1-A39 axis codes, marked APPROVED	Modules cannot enter clinical ranking until their axis mappings are expert-approved. We have prepared the candidate remap; it needs your sign-off.	No clinically usable module recommendations at all — the single biggest blocker.
2	Confirm each network's Primary/Secondary Axis (Blocker 2a) — sign off the crosswalk we sent	The 39-axis engine cannot use the 180 networks until each network's axis is confirmed. This also switches on the network-validation component of module scoring.	Network mechanistic validation stays off; module scores stay provisional.
3	Network connections / edges (Blocker 2c) — which network feeds which	Without edges, the AI cannot trace a problem to its root network or explain propagation.	The "network explanation" layer stays incomplete.
4	Clinical reference content — lab cut-off thresholds, per-claim evidence grades, and the per-axis minimum-evidence rules	These are the tables the engine reads instead of guessing. Without them, axes cannot be scored to a clinical standard and modules cannot be graded.	Axis scoring and module grading remain provisional, not production-grade.

Why this is time-sensitive: the engine is essentially finished waiting on *decisions* — it is now waiting on your *data and approvals*. These four items are sequential dependencies for the physician pilot. Starting the expert review now (even a few rows or one registry at a time) directly shortens the path to a working pilot. The corresponding fill-in files and the candidate mappings are already in your hands.

4c. Decision resolved (5 Aug) ✓

- **How the severity score treats incomplete evidence — now decided and implemented.** Your 5 Aug ruling is final: **data completeness must not reduce an observed abnormality's severity**. We reworked the severity score accordingly (**NSS v0.3**): a severe finding stays severe, and reliability (confidence, completeness, coverage, biological uncertainty) is reported alongside it rather than bent into the number. Only axes that reach the required assessment state contribute; provisional axes produce a clearly-labelled *Provisional* score. This item is closed.

5. Refinements completed in response to your 31 Jul guidance

We implemented your review points immediately. In plain terms:

1. **Ferritin** now reads as a blood/iron marker first (Axis A26) and only points to inflammation (A1) when inflammatory markers confirm it — correcting the earlier A34 typo.
2. **HOMA-IR** is calculated with the correct unit-specific formula and **refuses to compute if the unit is unknown**, preventing wrong insulin-resistance values.
3. **Each network now carries three separate identifiers** (a permanent code, a display number, and the legacy N-id), so future re-ordering never changes a network's identity.
4. **A network's Domain is derived automatically from its confirmed Axis** — no duplicate or contradictory domain data.
5. **Axis assessment uses your five official states** (Not Assessed → Axis Candidate → Provisionally Assessed → Assessed → High-Confidence Assessed).
6. **Modules with no graded evidence are marked "not yet graded" (neutral)** — never mislabeled as weakest.
7. **Safety is a clear categorical decision** (Pass / Pass-with-Monitoring / Hold / Contraindicated / Insufficient-Safety-Data), never a number.
8. **When network data is unavailable, the score shows "not assessed" plus a coverage percentage** — never a silent zero, and never quietly re-weighted.

Following your 5 Aug review we also: reworked the **severity score (NSS v0.3)** so completeness never reduces severity; recorded **CELA as approved** (TSPI-NET-0120 → A34 / D10); and locked the **A10 sub-axes** and the **nine Restoration Steps** to your canonical numbered lists. Your crosswalk / minimum-evidence corrections (delete the example row, add the missing fields, define the confidence scale, apply the semantic mapping fixes) remain queued for those two review spreadsheets.

6. The two blockers — where they stand and why they matter

Blocker 1 — Module Registry (● waiting on your approval). The official legacy → current axis mapping has been received, and our converter now automatically applies the clearly-safe mappings and **quarantines the risky ones for your review** (a blind number-swap is unsafe — e.g. legacy "Metabolic Balance" is *not* today's A27). **What remains is your approval of the candidate remap.** This is the highest-impact blocker: until it is approved, module recommendations cannot be used clinically. We can prepare the full candidate file for sign-off on request.

Blocker 2 — the 180-Network layer (● partially resolved). All 180 networks are transcribed and verified, **CELA is now approved** (TSPI-NET-0120 → A34), and we have built the axis/domain crosswalk for your review. **Three things remain:** (a) your **row-by-row confirmation of each network's Primary/Secondary Axis**, (b) the **network-to-network connections (edges)**, and (c) — per your 5 Aug review — a **Module-to-Network Claim Registry** (which module treats which network, with evidence). Note: the 25% network-match component depends on (c), not on the axis crosswalk alone; until it exists the component correctly stays *Not Assessed* rather than defaulting to zero.

In one line: the fastest path to a physician pilot is **(1) approve the Module Registry remap** and **(2) begin confirming the network axis mappings** — even incrementally. Both are ready for your review now.

7. Pre-pilot governance layer (added by your 5 Aug review)

Your final review introduced a set of governance and validation requirements that must exist before a physician pilot. None of them change what is already built; they define what must be in place before any clinical use. We are folding them into the implementation plan:

- **Module-to-Network Claim Registry** — the real basis for network match (see §6).
- **Evidence de-duplication** — one result (e.g. ferritin) must not score full weight into several axes at once.
- **Two-stage safety** — the pre-ranking eligibility gate we already have, **plus** a new regimen-level combination review (ingredient/mechanism duplication, additive effects, pill burden).
- **Statement-level provenance + an unsupported-statement detector** — every clinical sentence must trace to data and an approved rule, or be labelled a non-scoring hypothesis.
- **Ontology vs patient-assessment separation** · distinct **null / 0 / false / NOT_ASSESSED** semantics · a **persistent versioned Patient Biological Model** with explicit transition reasons.
- **Release-state ladder** (fail-closed; nothing provisional reaches patient release) · **cross-report consistency validator** · **registry dependency graph** · **rollback / incident management**.
- **Clinician-authored gold-standard test cases** and a **physician shadow-pilot** (no auto-prescribing, no patient release) — because the automated tests prove technical behaviour, not clinical validity.

Recommended sequence to a pilot (from your review): approve the Module remap → complete the network crosswalk → build the Network Edge Registry → build the Module-to-Network Claim Registry → approve the pilot clinical registries → lock NSS → run clinician gold-standard tests → physician shadow pilot → patient release only after formal approval.

8. Platform & access — how people use the AI Brain (new)

Alongside the clinical engine, we built the access layer so different people can use it safely.

Doctor's-chat integration. A doctor can now connect the AI Brain to their AI chat client and run the whole loop in plain language: **submit a de-identified case → screen for red flags → generate a draft plan → approve or update it.** The chat is only a front-end; the engine remains the sole source of clinical conclusions, and **no patient identifiers ever reach it.** (Validated end-to-end.)

Approve and update. Beyond approve/reject, a clinician can now make **structured, reason-coded edits** to a plan (remove a module, change a dose, override safety with justification, adjust axes). Every edit is recorded, and editing an approved plan sends it back to draft for re-approval.

Roles & access (as you requested). The system now enforces roles end-to-end:

Role	Can do
Patient	self-register, upload own labs, submit symptoms; view only their own approved plan
Clinic staff	everything a patient can, plus trigger generation, view live status, and view all patients' reports in their clinic — no clinical approval/override
Clinician	generate, update, and approve/reject their own cases
Reviewer (Clinical Review Board)	the above, plus govern the learning model
Auditor	read-only access to reports + the audit trail

Patient identity stays in MiHealth; the engine only ever sees a de-identified case. Every clinically meaningful action is written to an **append-only audit trail** (who, what, when, why). Access control ships **switched off by default** and is turned on per environment before multi-user or real-patient use.

Two decisions we'd like your confirmation on: 1. For an **unapproved draft**, should clinic staff see the full clinical detail, or only its **status + safety flags** until a clinician approves it? (*We recommend status + safety flags only.*) 2. Confirm that **patient and clinic-staff sign-in lives in MiHealth** (the patient portal), with the approved plan shown to the patient there — the AI-chat path stays clinician-only.